

YILIN WANG

Senior Staff AI Scientist · Multimodal Foundation Models · Post-Training & RL · Data and Evaluation at Scale

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yilinwang.org · Google Scholar · LinkedIn · GitHub

8 PRODUCTION TRANSFERS · 55+ PAPERS, ~2,700 CITATIONS · 13 INTERNS MENTORED · 46 U.S. PATENTS

Research scientist working at the intersection of post-training, large-scale data, and evaluation for multimodal generative models. Seven years building frontier-adjacent systems that ship: eight research-to-production transfers across Photoshop, Lightroom, and Firefly, including the instruction-editing stack in Firefly Image 5. Publishes and productionizes on the same loop — RL post-training methods, multi-million-example data pipelines, and the evaluation infrastructure that tells you whether either one worked.

CORE EXPERTISE

Post-training & RL: GRPO and policy-gradient methods for generative models; reward design and credit assignment; preference data; diffusion and flow matching; teacher/student distillation; RLHF-style human feedback loops.

Data & evaluation: Multi-million-example curation and filtering; synthetic and video-derived supervision; multimodal annotation pipelines; human and VLM-as-judge evaluation; benchmark design and failure taxonomies.

Multimodal modeling: VLM/MLLM training and planning systems; multimodal representation learning; image generation and instruction-based editing; segmentation, matting, and style transfer.

Systems & leadership: Python, PyTorch, distributed training; production deployment at consumer scale; cross-functional execution; research mentorship.

PROFESSIONAL EXPERIENCE

Adobe — Senior Staff AI Scientist

San Jose, CA · 2019 – Present

- **Firefly Image 5 — instruction editing.** Core contributor across data strategy, teacher-model training, evaluation, and distillation for the instruction-editing stack, enabling precise natural-language edits that preserve subject identity and image quality in production.
- **RL post-training for flow-matching models.** Co-developed Stepwise-Flow-GRPO, which attributes reward improvement to individual denoising steps rather than whole trajectories, producing denser and more informative credit assignment for generative post-training (CVPR 2026). Drove reward design, preference-data collection, and distillation of post-trained teachers into deployable student checkpoints.
- **Data systems at scale.** Co-led multi-million-example paired-editing and instruction-data efforts: video-derived supervision, caption-to-instruction synthesis, automated annotation, and multi-stage quality filtering. The same filtering methodology produced OmniStyle-1M, a one-million-pair curated style-transfer corpus released with the accompanying paper.
- **Evaluation infrastructure.** Built automated, VLM-based, and human evaluation workflows for instruction following, edit fidelity, visual similarity, and compositional understanding; converted failure taxonomies into targeted data and training interventions. Introduced relational visual similarity, a relationship-centered metric complementing LPIPS, CLIP, and DINO (CVPR 2026).
- **Vision foundation systems.** Initiated and delivered the selection and refinement models behind Select Subject, Object Finder, and Select People Details — one-click, high-precision segmentation shipped in Photoshop and Lightroom and used daily by millions of creators.
- **Research leadership.** Mentored 13 research interns and external collaborators, serving as corresponding author or co-advisor on work published at CVPR, ICLR, NeurIPS, ECCV, and AAAI. Named inventor on 46 granted U.S. patents. Several mentees now hold research positions at frontier labs and top-tier faculty appointments.

SHIPPED SYSTEMS

Eight research-to-production technology transfers across Photoshop, Lightroom, and Firefly. Selected:

- **Firefly Image 5.** Instruction-based image generation and editing — precise, natural-language-driven edits at production quality.
- **Select People Details (Photoshop).** Automatic detection and selection of individuals and fine-grained regions — hair, skin, clothing, facial features, hands — enabling precise individual and group editing.

- **Object Finder (Photoshop)**. Automatic identification and selection of every object in an image — people, items, sky — shipped as part of the Object Selection Tool.
- **Select Subject (Photoshop, Lightroom)**. One-click isolation of an image's primary subject; fast, accurate selections for photographers and designers.

SELECTED PUBLICATIONS

Post-Training & Reinforcement Learning for Generative Models

- Stepwise Credit Assignment for GRPO on Flow-Matching Models · **CVPR 2026**
- UniReal: Universal Image Generation and Editing via Learning Real-World Dynamics · **CVPR 2025** (*Highlight*)
- Incorporating Reinforced Adversarial Learning in Autoregressive Image Generation · **ECCV 2020**

Multimodal Understanding, Planning & Evaluation

- Beyond Simple Edits: X-Planner for Complex Instruction-Based Image Editing · **AAAI 2026**
- Relational Visual Similarity · **CVPR 2026**
- FineCaption: Compositional Image Captioning · **CVPR 2025**
- Multimodal Contrastive Training for Visual Representation Learning · **CVPR 2021**

Data-Centric Generation & Personalized Editing

- OmniStyle2: Learning to Stylize by Learning to Destylize · **ECCV 2026** (*corresponding author, project lead*)
- OmniStyle: Filtering High-Quality Style Transfer Data at Scale · **CVPR 2025** (*corresponding author*)
- SwapAnything: Arbitrary Object Swapping in Personalized Visual Editing · **ECCV 2024** (*co-advisor*)
- PhotoSwap: Personalized Subject Swapping in Images · **NeurIPS 2023**
- UniHuman: A Unified Model for Editing Human Images in the Wild · **CVPR 2024**

Vision Foundations

- Amodal Scene Analysis via Holistic Occlusion Relation Inference and Generative Mask Completion · **AAAI 2024** (*Oral*)
- Lite Vision Transformer with Enhanced Self-Attention · **CVPR 2022**
- Mask Guided Matting via Progressive Refinement Network · **CVPR 2021**

Full list (55+ papers, ~2,700 citations) at yilinwang.org and [Google Scholar](https://scholar.google.com/citations?user=...).

EDUCATION

Ph.D., Computer Science — Arizona State University, 2018. Advisor: Baoxin Li.

Dissertation research in multimodal learning and visual understanding, with publications at CVPR, NeurIPS, IJCAI, AAAI, WWW, and WSDM.

[Add M.S. and B.S. — institution, field, year]

SERVICE & COMMUNITY

Area Chair: ACM Multimedia 2020, 2021.

Reviewer: CVPR, ICCV, ECCV, ICML, NeurIPS (since 2017).

Mentorship: 13 research interns hosted at Adobe, plus ongoing external co-advising with academic collaborators.